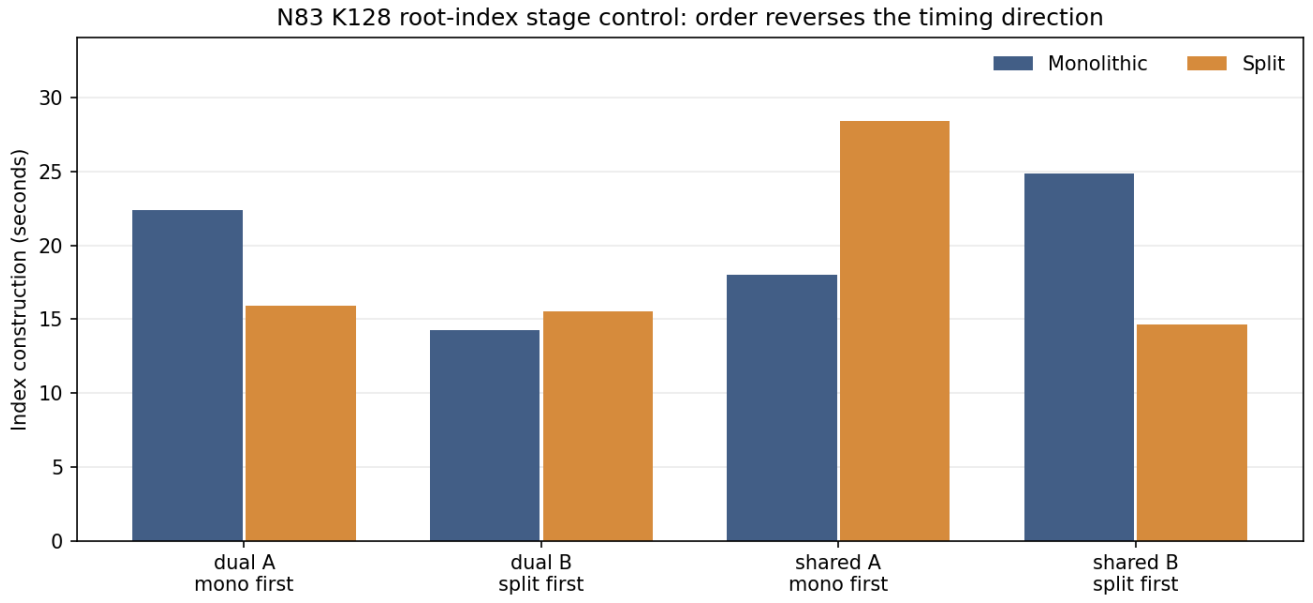
N83 K128 split root index: result

The split exact-root construction preserved the K128 base, state set, and root-entry count. Index-build timing changed direction with run order: the paired local timings do not establish a stage speedup.

At K128 the monolithic five-byte table allocated 10,485,760 bytes and the split tables allocated 13,107,200 bytes. Shared Bloom allocated 2,097,152 bytes, matching monolithic and saving 524,288 bytes compared with separate split-table filters. The frozen K4405 layout formula saves 128 MiB of Bloom space against separate filters and adds 671,088,640 table bytes against monolithic.

Status: STAGE_DIAGNOSTIC_ONLY. Candidate ID and single-target online speedup remain unset. A full-size K4405 same-target scalar replay and Tier 1 paired measurement are the next qualification steps.

Order	Mode	Mono (s)	Split (s)	Faster stage
k128_a	dual	22.393	15.919	split
k128_b	dual	14.285	15.502	mono
k128_shared_a	shared	18.030	28.390	mono
k128_shared_b	shared	24.862	14.632	split



Source: research/n83_two_tier_root_20261021/runs/{k128_a,k128_b,k128_shared_a,k128_shared_b}/result.json; same study directory contains inputs, stderr, stdout, and runner copies.